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Faculty of Computers and Artificial
Intelligence



Program: General

IS-211 Introduction to DB Systems

Project Title: Pharma Sales Data
Project Phase 1

Section: S11,12

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Pharma Sales Data

Step 1: Source Your Data:

Dataset Name: Pharma Sales Data

- The dataset chosen for this project is the Pharmaceutical Sales Data published on Kaggle by Milan Zdravkovic. It contains transactional sales records from a real community pharmacy in Bosnia and Herzegovina, capturing unit sales for 8 drug categories at four levels of granularity: hourly, daily, weekly, and monthly.
- Dataset link: [Pharma Sales Data – Kaggle](#)

Step 2: Understand and Explain the Dataset:

The dataset is built from the initial dataset consisted of 600000 transactional data collected in 6 years (period 2014-2019), indicating date and time of sale, pharmaceutical drug brand name and sold quantity, exported from Point-of-Sale system in the individual pharmacy. Selected group of drugs from the dataset (57 drugs) is classified to the following Anatomical Therapeutic Chemical (ATC) Classification System categories:

- M01AB - Anti-inflammatory and antirheumatic products, non-steroids, Acetic acid derivatives and related substances
- M01AE - Anti-inflammatory and antirheumatic products, non-steroids, Propionic acid derivatives
- N02BA - Other analgesics and antipyretics, Salicylic acid and derivatives
- N02BE/B - Other analgesics and antipyretics, Pyrazolones and Anilides
- N05B - Psycholeptics drugs, Anxiolytic drugs
- N05C - Psycholeptics drugs, Hypnotics and sedatives drugs
- R03 - Drugs for obstructive airway diseases
- R06 - Antihistamines for systemic use

Sales data are resampled to the hourly, daily, weekly and monthly periods. Data is already pre-processed, where processing includes outlier detection and treatment and missing data imputation.

Where did the data come from?

- The data originates from a single community pharmacy located in Bosnia and Herzegovina. Sales records were collected across all business hours over a 6-year window (January 2014 through September 2019). The dataset was published on Kaggle by Milan Zdravkovic and is freely available for research and educational use. The drugs are classified using the Anatomical Therapeutic Chemical (ATC) system - the internationally recognised WHO standard for classifying medications.

What does each row represent?

Table	Each Row Represents
saleshourly	One hour of sales on a specific date and time (e.g., Thursday 2016-03-10 at 10:00 AM) (50,532 rows)
salesdaily	One full calendar day of aggregated drug sales (e.g., all sales on Monday, 15 July 2016) (2,106 rows)
salesweekly	One ISO calendar week of aggregated drug sales (e.g., the week starting 5 January 2014) (302 rows)
salesmonthly	One complete calendar month of aggregated drug sales (e.g., all sales in October 2016) (70 rows)

What do the key columns mean?

The table below defines all columns present across the four tables. Drug codes follow the WHO ATC (Anatomical Therapeutic Chemical) classification system.

Column	Data Type	Full Description
datum	Date / Datetime	Primary key. Observation timestamp. Format varies by table: date for daily/weekly/monthly; datetime (YYYY-MM-DD HH:MM) for hourly.
M01AB	Float	Non-steroidal anti-inflammatory drugs (NSAIDs) - acetic acid and related substances. Used for pain and inflammation.
M01AE	Float	NSAIDs - propionic acid derivatives (e.g. Ibuprofen group). Second largest anti-inflammatory category.
N02BA	Float	Salicylic acid-based analgesics and antipyretics (e.g. Aspirin group). Used for pain relief and fever.
N02BE	Float	Anilide analgesics and antipyretics - primarily Paracetamol (Acetaminophen). THE top-selling drug category at 49.4% of all sales.
N05B	Float	Central nervous system drugs used to reduce anxiety. Second-highest volume drug category.
N05C	Float	CNS drugs that induce or maintain sleep. Lowest volume category in the dataset.
R03	Float	Respiratory drugs for asthma, COPD, and bronchospasm - includes inhalers and bronchodilators.
R06	Float	Respiratory antihistamines for allergy treatment. Peaks in spring/summer months.
Year	Integer	Calendar year extracted from datum. Present in daily and hourly files.

Month	Integer	Calendar month (1 = January ... 12 = December). Present in daily and hourly files.
Hour	Integer	Hour of observation on a 24-hour clock. Only meaningful in hourly and daily files.
Weekday Name	String	Full name of the weekday for the observation date. Enables weekday demand pattern analysis.

Step 3: Define the Problem (The "Why")

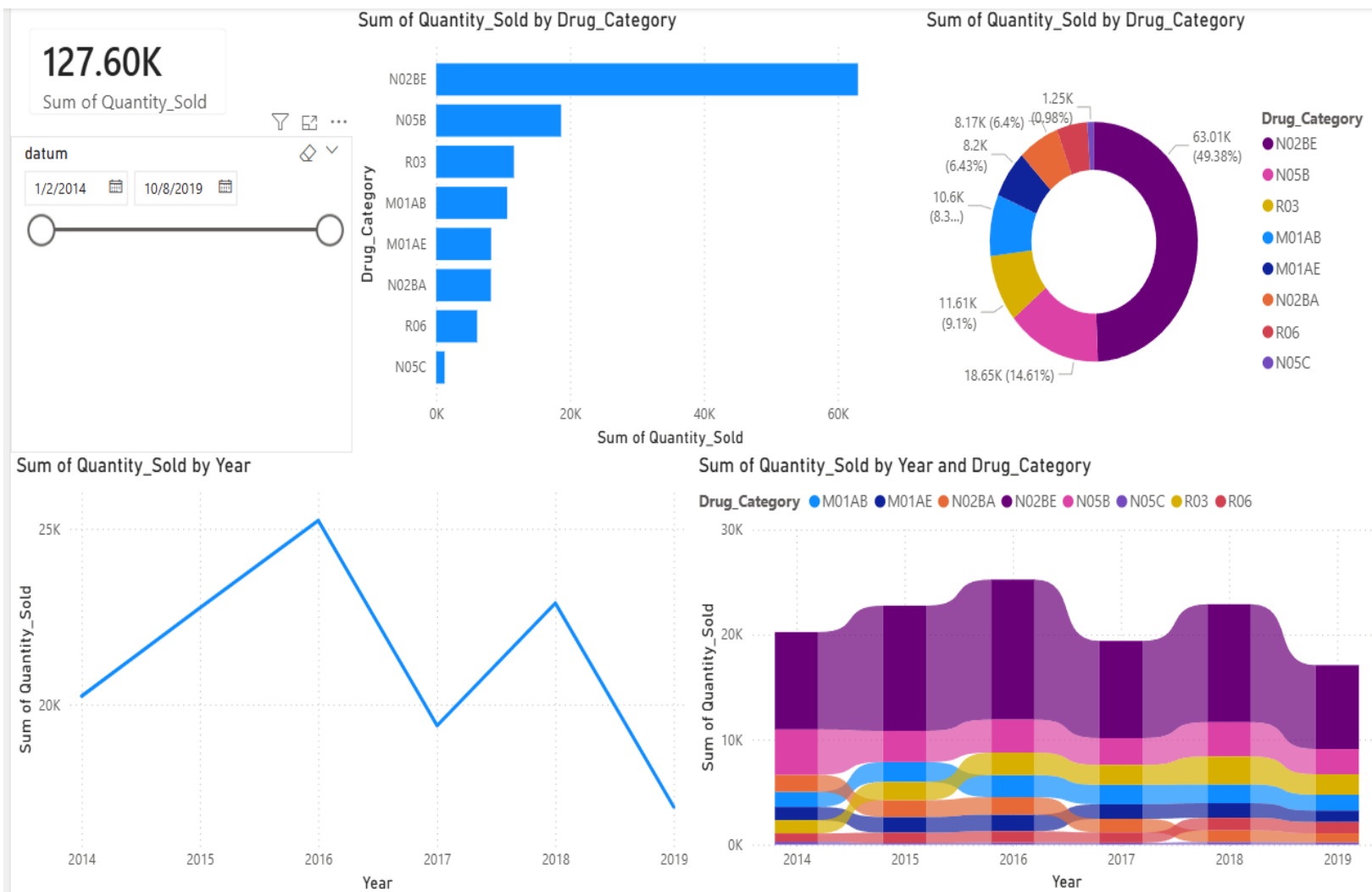
Who is the user?

- The user is a **Pharmacy Operations Manager**.

What is the problem?

- The pharmacy manager currently lacks a consolidated, visual view of sales performance. The manager cannot quickly identify which drug categories are driving revenue, whether sales follow predictable seasonal patterns, or which months/days underperform, making stock planning and staff scheduling inefficient.

Step 4: Build the Interactive Dashboard (The "What")



Step 5: Create the Detailed Report (The "Why")

Annual Sales Summary (All Drug Categories Combined)

Year	M01AB	M01AE	N02BA	N02BE	N05B	N05C	R03	R06	Annual Total	YoY Change	YoY %
2014	1,447.2	1,257.1	1,616.5	9,274.4	4,315.1	272.6	1,255.0	800.5	20,238.1	N/A	N/A
2015	1,895.6	1,496.6	1,608.6	11,923.2	2,965.0	159.0	1,721.3	983.0	22,752.1	2,514.0	12.4%
2016	2,107.3	1,589.4	1,655.2	13,335.6	3,138.3	207.0	2,137.0	1,065.1	25,235.1	2,482.6	10.9%
2017	1,846.6	1,387.3	1,288.3	9,258.8	2,555.5	180.3	1,893.6	988.9	19,399.1	-5,835.6	-23.1%
2018	1,786.9	1,389.7	1,123.8	11,230.9	3,266.2	235.0	2,655.3	1,196.8	22,885.1	3,485.2	18.0%
2019	1,517.3	1,084.5	879.8	7,982.5	2,405.6	196.0	1,946.7	1,073.6	17,086.1	-5,798.6	-25.3%
TOTAL	10,601.1	8,205.1	8,172.1	63,005.1	18,646.1	1,250.1	11,609.1	6,108.1	127,596.1		

The 2017 Sales Drop: Monthly Drill-Down (Investigated Trend)

Total 2017 sales dropped 23.1% vs 2016 peak (25,235 → 19,393 units). Report isolates which months and drug categories drove this collapse.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Full Year
2016	2,451.1	2,232.1	2,235.1	2,148.1	1,957.1	1,725.1	1,600.1	1,633.1	2,050.1	2,655.1	1,943.1	2,606.1	25,235.1
2017	2,700.1	1,447.1	1,377.1	1,342.1	1,369.1	1,307.1	1,251.1	1,302.1	1,661.1	2,050.1	1,706.1	1,882.1	19,393.1
Change	248.1	-785.1	-858.1	-806.1	-588.1	-418.1	-349.1	-331.1	-389.1	-605.1	-237.1	-724.1	-5,841.1

The 2017 Sales Drop: Monthly Drill-Down (Investigated Trend)

Drug Category	Full Name	2014	2015	2016	2017	2018	2019	2016→2017 Drop	% Change	Trend
M01AB	Anti-inflammatory (Acetic ac	1,447.1	1,896.1	2,107.1	1,847.1	1,787.1	1,517.1	-261.1	-12.4%	↓ Drop
M01AE	Anti-inflammatory (Propionic	1,257.1	1,497.1	1,589.1	1,387.1	1,390.1	1,085.1	-202.1	-12.7%	↓ Drop
N02BA	Analgesics (Salicylic acid)	1,616.1	1,609.1	1,655.1	1,288.1	1,124.1	880.1	-367.1	-22.2%	↓ Major Drop
N02BE	Analgesics/Antipyretics (Pyr	9274.1	11923.225	13335.58	9258.8048	11230.925	7982.491	-4,077.1	-30.6%	↓ Major Drop
N05B	Psycholeptics - Anxiolytics	4,315.1	2,965.1	3,138.1	2,556.1	3,266.1	2,406.1	-583.1	-18.6%	↓ Drop
N05C	Psycholeptics - Hypnotics/S	273.1	159.1	207.1	180.1	235.1	196.1	-27.1	-12.9%	↓ Drop
R03	Drugs for Airway Diseases	1,255.1	1,721.1	2,137.1	1,894.1	2,655.1	1,947.1	-243.1	-11.4%	↓ Drop
R06	Antihistamines	800.1	983.1	1,065.1	989.1	1,197.1	1,074.1	-76.1	-7.2%	↓ Drop

Average Daily Sales by Day of Week (All Years)

Day of Week	Avg Daily Sales	Total Sales (All Years)	# of Trading Days
Saturday	65.67	19,768.1	301
Sunday	61.13	18,401.1	301
Monday	60.61	18,243.1	301
Friday	60.25	18,135.1	301
Tuesday	60.02	18,065.1	301
Wednesday	59.24	17,772.1	300
Thursday	57.18	17,212.1	301

Step 6: Present Your Insights

1. Make The Most Of Inventory and Resources For Category N02BE

The numbers show that category N02BE is the reason you sell a lot of products. It makes up 49.38% of everything you sell. Category N02BE always has the sales numbers every year.

Action: Focus on managing the supply chain and inventory for N02BE so you never run out of stock. Since category is almost half of your total sales any problems with supply will affect your money a lot. You should think about keeping more of category N02BE in stock to be safe in case a lot of people want to buy it or if there are problems, with getting more. This way you can make sure you have category N02BE when you need it. Category N02BE is very important so you need to take care of it.

2. Analyze the 2019 Performance Drop

The line chart shows a drop in total quantity sold in 2019 which breaks the trend we saw from 2014 to 2018. We need to take a look at the 2019 operational data right away. We have to check if the decline was due to problems outside of our control or if it was caused by issues with our operations.

It is crucial to find out if this was a one-time problem or if the market is actually shrinking. This will help us plan for the fiscal period. We must investigate the quantity sold in 2019. Understand what led to the decline, in quantity sold.